Error-Correcting Codes

Alice $\xrightarrow[\text{flip} \leq 1 \text{ bit}]{\text{Noisy channel}}$ Bob

101
↓
111000111 ⟶ 101000111
1 0 1

rate of transmission: $\frac{k}{3k} = \frac{1}{3}$

To correct t bits, one may repeat each bit $(2t+1)$ times

$A :=$ alphabet A^k: message
$A^n :=$ codes

Coding Scheme: $\begin{cases} \text{Encoding fct } f: A^k \longrightarrow C \subseteq A^n \\ \text{Decoding fct } g: A^n \longrightarrow A^k \end{cases}$

rate of transmission: $\frac{k}{n}$

__Hamming__: $\text{dist}(x,y) :=$ # of pos of different entries

$\text{dist}(C) = \min_{x,y \in C} \text{dist}(x,y)$

Hamming Ball: $B_d(x) = \{y \in \{0,1\}^n : \text{dist}(x,y) \leq d\}$

Given $x \in \{0,1\}^n$

__Linear Code__: $C \subseteq \{0,1\}^n$ is a linear code if C is a linear space

- $\forall x, y \in C \quad x+y \in C$

Let $k = \text{rank}(C) \quad |C| = 2^k$

__Generator__: $G \subseteq \mathbb{F}_2^{k \times n}$ rows form a basis

Parity-Check: $H \subseteq \mathbb{F}_2^{(n-k) \times n}$ whose rows form basis of $C^{\perp} = \{y \in \mathbb{R}^n : \langle y, x \rangle = 0\}$ $C = \{x \mid Hx = 0$

Messages: $u \in \{0,1\}^k \quad y = G^T u \in C \xrightarrow[\leq t]{\text{Noise}} y'$

Find unique $y \in B_t(y')$ s.t $Hy = 0$

$\equiv$

$a \in B_t(0)$ s.t $Hy' = Ha \Leftrightarrow H(y'+a) = H(y'-a)$

$y = y' + a$, a tells ya where errors happen

__Reed-Solomon Codes:__

$A = \mathbb{F}_q = \{0, \ldots q-1\}$

messages: $\mathbb{F}_q^k \quad W = (W_0, \ldots, W_{k-1})$

$P_W(x) = W_0 + W_1 x + \cdots + W_{k-1} x^{k-1}$

Distinct $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q \quad q \geq n$

$f(W) = (P_W(\alpha_1), \ldots P_W(\alpha_n)) \in \mathbb{F}_q^n$

$C = \{f(w) \mid w \in \mathbb{F}_q^k\}$ $\text{dist}(C) \geq n-k+1$

These are mathematical garuntees, but how to actually find the associated codeword for the errored one

- Hamming Code: Syndrome Testing
- Reed-Solomon: Polynomial reconstruction
- LDPC: Belief propagation
- Turbo Codes: Iterative Decoding